

Dwaipayan Dasgupta

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SUMMARY

AI/ML undergraduate specializing in Machine Learning, Deep Learning and Artificial Intelligence. Dedicated to transforming data into meaningful insights using modern programming languages and frameworks. Passionate about building impactful AI-driven solutions with proven problem-solving and team collaboration skills to support successful product development.

EDUCATION

Heritage Institute of Technology <i>B.Tech in Computer Science Engineering (AI/ML)</i>	Kolkata, WB 2023 – 2027
Delhi Public School, Joka, South Kolkata <i>Central Board of Secondary Education (CBSE)</i>	Kolkata, WB 2021 – 2023
Vivekananda Mission School, Joka <i>Indian Certificate of Secondary Education (ICSE)</i>	Kolkata, WB 2021

PROJECTS

StockSense – Stock Analysis System | *Machine Unlearning, FinTech* | [GitHub](#)

- Built a stock analysis web-application that ingests data, features and fine-tunes Yi-6B/Qwen-1.0B LLM models to predict short-term stock price movements using AAPL datasets.
- Implemented machine unlearning techniques using gradient ascent and descent to prevent data poisoning and update prediction models efficiently without complete model retraining.
- Developed adaptive model pipelines to maintain accurate, compliant and continuously updated financial predictions.

CarbonTrack | *AI/ML, Data Analytics* | [GitHub](#)

- Developed an AI-driven carbon tracking system for coal mines to forecast emissions and recommend solutions for achieving carbon neutrality.
- Designed a predictive LSTM model to estimate carbon footprints of mining operations using satellite imagery and operational parameters.
- Incorporated sustainability factors including forest coverage, mine location and industrial activity for comprehensive environmental impact analysis and carbon neutrality planning.

SalesAI – Conversational Loan System | *Agentic AI, FinTech* | [GitHub](#)

- Architected a multi-agent AI orchestration system to automate the complete loan lifecycle, including customer interaction and sanction letter generation.
- Built a technical underwriting engine using financial formulas and risk analysis for loan pre-approval and credit eligibility assessment.
- Improved loan processing efficiency through conversational AI and automated decision-making workflows.

Pillchill – Centralized Ecosystem for Healthcare | *Machine Learning, Healthcare* | [GitHub](#)

- Developed a personal health dashboard with editable vitals and integrated tracking for appointments, medications and test reports.
- Built an OCR pipeline to automate prescription and medical data extraction with high accuracy in dosage and frequency recognition.
- Implemented real-time appointment scheduling and community health content sharing features to enhance patient engagement and awareness.

TECHNICAL SKILLS

Languages: Python, R, C/C++, SQL, Java, MATLAB

Frameworks: PyTorch, TensorFlow, Keras, LangChain, MLflow, Apache Spark, Kubernetes

Developer Tools: Git, GitHub, MongoDB, PostgreSQL, Docker, Weka

Leadership Skills: Team Leadership, Time Management, Problem Solving, Critical Thinking, Effective Communication